NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

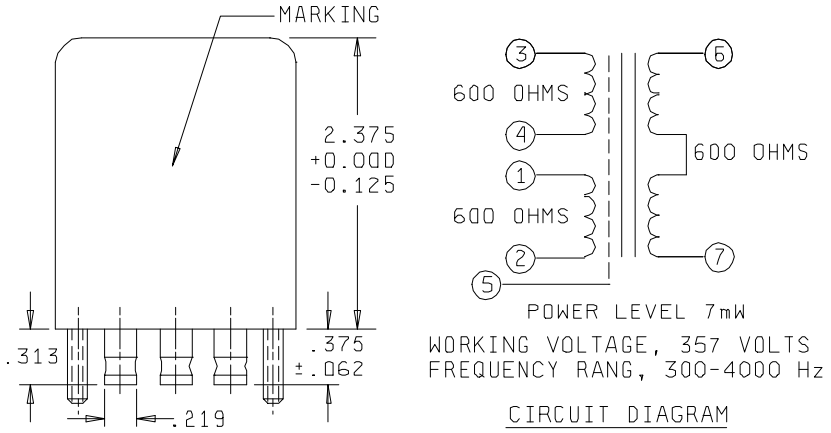
MIL-PRF-27/307
6 FEBRUARY 1981

PERFORMANCE SPECIFICATION SHEET

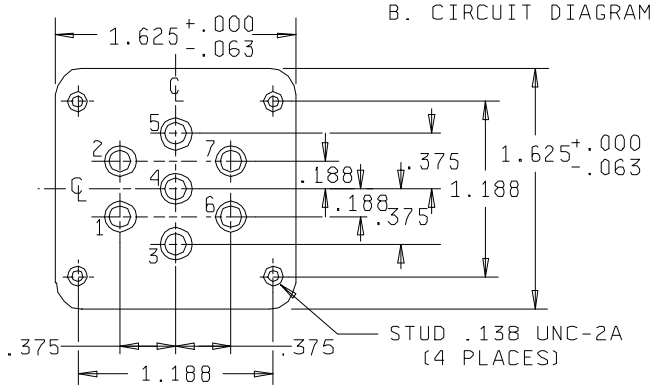
TRANSFORMER, AUDIO FREQUENCY, TF4S21AJ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of Specification MIL-T-27.



B. CIRCUIT DIAGRAM AND MARKING



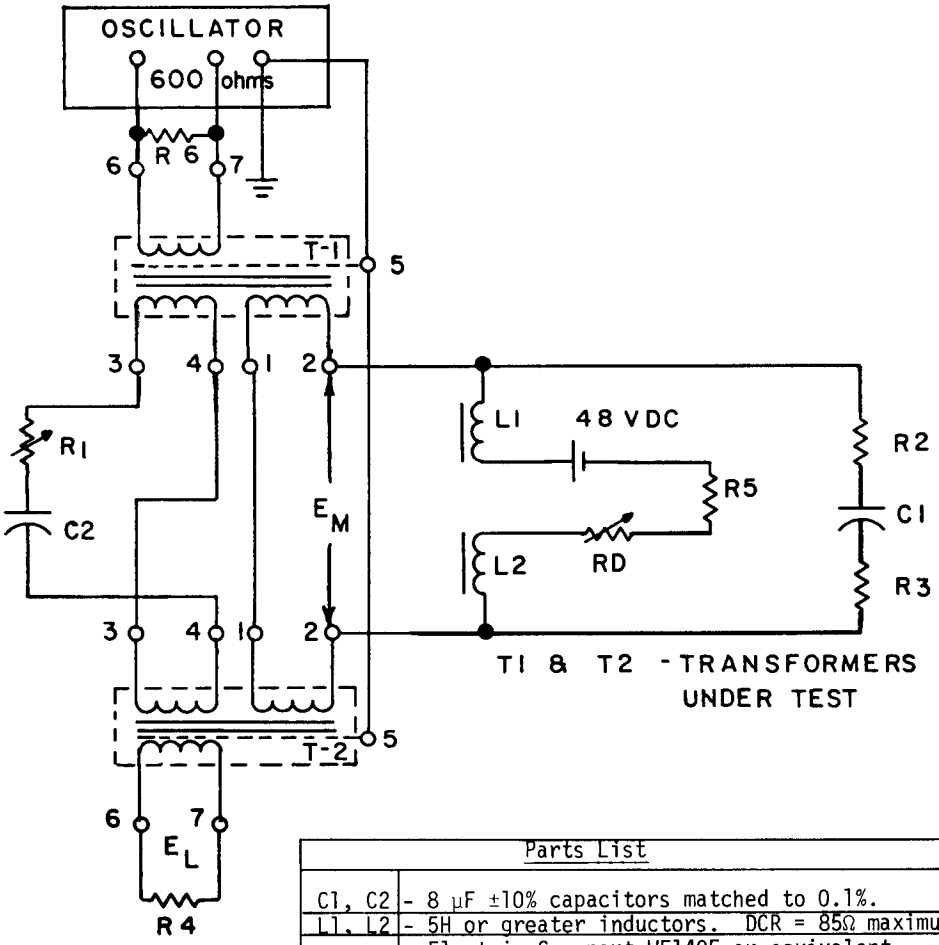
A. PHYSICAL DIMENSIONS

INCHES	MM
.063	1.60
.125	3.18
.188	4.78
.219	5.56
.313	7.95
.375	9.52
1.188	30.18
1.625	41.27
2.375	60.32

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified the tolerance shall be ±.016 inch (0.41 mm).

FIGURE 1. Dimensions and configurations

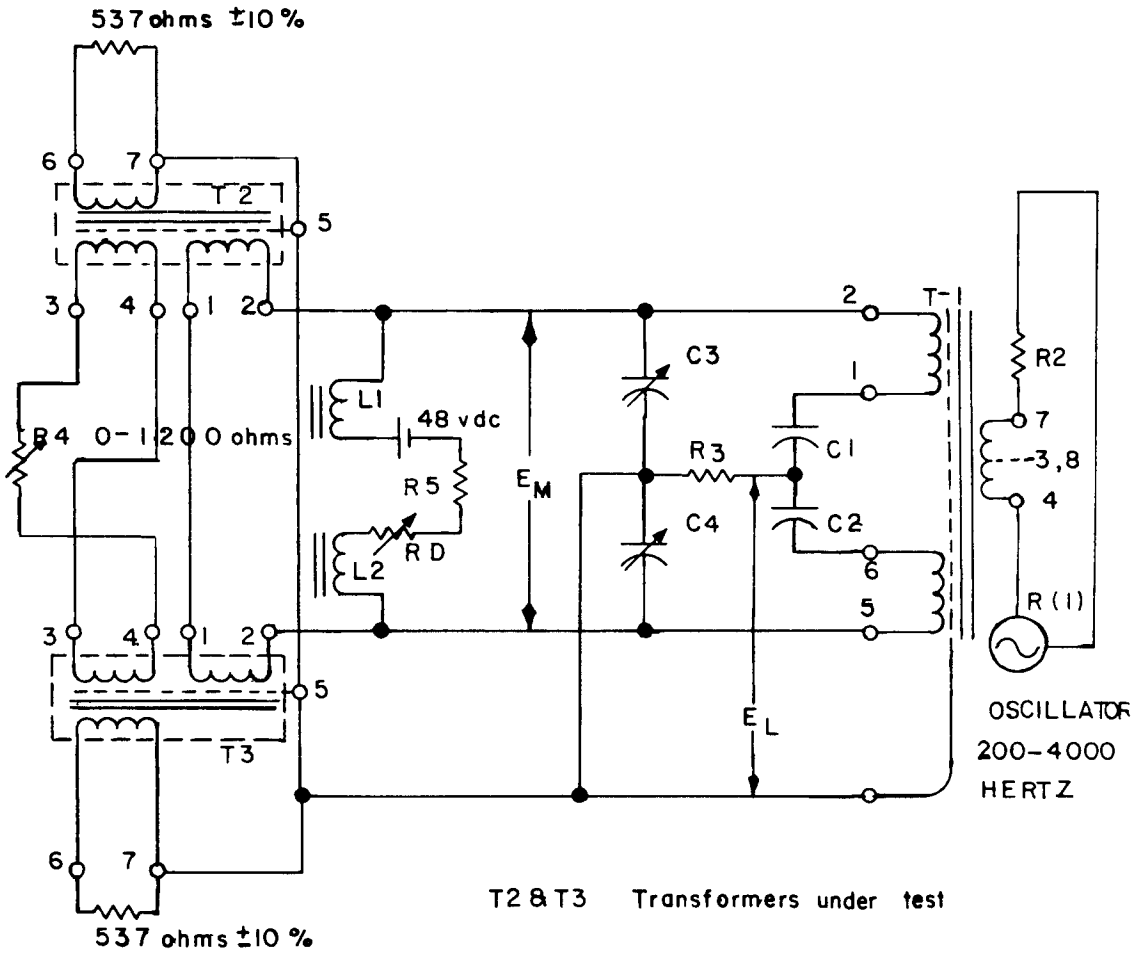
Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:07Z
Check the source to verify that this is the current version before use.



Parts List	
C1, C2	- 8 µF ±10% capacitors matched to 0.1%.
L1, L2	- 5H or greater inductors. DCR = 85Ω maximum Western Electric Co. part WE149E or equivalent.
R2, R3	- Resistors, such that R2 + R3 = 600Ω ±1%.
RD	- Variable resistor 0-1200Ω.
R1	- Variable resistor 0-1200Ω.
R4	- Resistor, 537Ω ±1%.
R5	- Resistor, 300Ω ±10%.
R6	- Resistor, 5110Ω ±10%.

FIGURE 2. Test circuit for transhybrid loss.

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Parts List	
T1	- Transformer, Western Electric Co. part WE111C or WE93A or ADC part 118F or equivalent.
C1, C2	- 8 µF ±10% capacitor matched to 0.1%.
C3, C4	- 100 to 500 pF adjustable trimmer capacitors.
OSC	- Audio oscillator with source resistance R1 = 600Ω.
R2	- Resistor such that R1 + R2 = 600Ω.
R3	- Resistor, 500Ω ±10%.
L1, L2	- 5H or greater inductor. DCR = 85Ω maximum Western Electric Co. part WE149E or equivalent.
RD	- Variable resistor, 0-1200Ω. Adjust for DC loop current of 100 mA.
R4	- Variable resistor, 0-1200Ω.
R5	- Resistor, 300Ω ±10%.

FIGURE 3. Test circuit for longitudinal balance.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the extreme terminals of the winding.)

Electrical ratings:

Power level: 7 mW.

Working voltage:

(1-2): 357 V maximum.
(3-4): 357 V maximum.
(6-7): 357 V maximum.

Frequency range: 300 Hz to 4,000 Hz.

Line winding current (1-2): 100 mAdc.

Line winding impedance (1-2): 600 ohms.

Four wire winding impedance (6-7): 600 ohms.

Balance winding impedance (3-4): 600 ohms.

DC resistance:

(1-2): 23 ohms ±15%.
(3-4): 23 ohms ±15%.
(6-7): 47 ohms ±15%.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Metal encased.

Material: Steel.

Terminals: Standoff, solder type.

Height: .313 inch.

Weight: 325 grams maximum.

Operating temperature range: -55° to +85°C.

Altitude: 10,000 feet maximum.

Terminal strength: Method 211 of MIL-STD-202, test condition A, 5 pounds.

Dielectric withstanding voltage: Method 301 of MIL-STD-202.

Test voltage: 1,000 Vrms.

Moisture resistance: Method 106 of MIL-STD-202, 100 Vdc polarizing voltage; subcycles 7A and 7B are not applicable.

Shock (specified pulse): Method 213 of MIL-STD-202, test condition I.

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Electrical characteristics:

Harmonic distortion: Not more than .5% at +7 dBm at 1 kHz.

Line winding impedance (1-2): 600 ohms ±10%.

Four wire winding impedance (6-7): 600 ohms ±10%.

Balance winding impedance (3-4): 600 ohms ±10%.

DC resistance:

(1-2): 23 ohms ±15%.
(3-4): 23 ohms ±15%.
(6-7): 47 ohms ±15%.

Frequency response: 300 Hz to 4000 Hz, flat within ±0.3 dB, reference frequency 1 kHz using test circuit shown on figure 2.

Insertion loss: 5 ±1 dB with 1 Vrms at 1 kHz applied across (3-5) with (3-4) and (1-2) connected in series.

Polarity: Additive, with terminals 1 and 4 and 2 and 6 connected together.

Transhybrid loss: 40 dB minimum when tested in accordance with figure 2 and as follows: (NOTE: Two transformers are required to perform this test.)

a. Adjust RD for a loop current of 100 mA.
b. Adjust oscillator for E_M = .775 volts at 1000 Hz.
c. Adjust R1 for E_L minimum.
d. Compute transhybrid loss = 20 log $\frac{E_{6-7}\ (T1)}{E_L\ (T2)}$.

Longitudinal balance: 40 dB minimum when tested in accordance with figure 3 and as follows:

a. Replace all equipment to the left of E_M with a 600-ohm resistor.

b. Adjust oscillator for E_M = .775 volts at 1000 Hz.

c. Adjust C3 and C4 for ratio 20 log $\frac{E_M}{E_L} \geq$ 60 dB.

d. Replace 600-ohm resistor with the rest of the test circuit as shown on figure 3.

e. Readjust oscillator for E_M = .775 volts.

f. Adjust RD for loop current of 100 mA dc.

g. Adjust R4 to 600 ohms.

h. Compute longitudinal balance = 20 log $\frac{E_M}{E_L}$.

Marking location: See figure 1.

Part number: M27/307-01.

QUALITY ASSURANCE PROVISIONS:

Qualification inspection: Transhybrid loss and longitudinal balance tests shall be performed as part of the electrical characteristics tests for group II inspections.

Quality conformance inspection: Transhybrid loss and longitudinal balance tests shall be performed as part of the electrical characteristics tests for group A, subgroup II inspections.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME, MU, WC
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5950-0568-24)

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